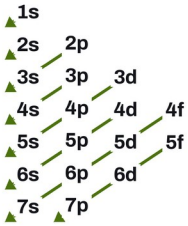
PERIODIC TABLE BY SUBLEVEL

UNIT 2 TEST · REFERENCE · DO NOT WRITE ON THIS SHEET

	s-BLOCK		d-BLOCK										p-BLOCK					
group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
valence e⁻	1	2	valence = the electrons in the outer s sublevel										3	4	5	6	7	8 He: 2
1s	1 H																	2 He
2s	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3s	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4s	19 K	20 Ca	3d 21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	4p 31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5s	37 Rb	38 Sr	4d 39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	5p 49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6s	55 Cs	56 Ba	57–70 4f ↓ 5d 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	6p 81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7s	87 Fr	88 Ra	89–102 5f ↓ 6d 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	7p 113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og

DIAGONAL RULE



FILLING ORDER, ROW BY ROW

Row 1	1s
Row 2	2s 2p
Row 3	3s 3p
Row 4	4s 3d 4p
Row 5	5s 4d 5p
Row 6	6s 4f 5d 6p
Row 7	7s 5f 6d 7p

SUBLEVEL CAPACITY

	ORBITALS	MAX e⁻
s	1	2
p	3	6
d	5	10
f	7	14

f-BLOCK · fits between groups 2 and 3 in rows 6 and 7

LANTHANIDES	4f	57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb
ACTINIDES	5f	89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No

READING A ROW

Read left to right and say the sublevel labels as you pass them. Row 4: 4s, then 3d, then 4p. The d sublevel is always one level lower than the row; f is two lower.

NOBLE-GAS SHORTHAND

Start from the noble gas at the end of the row above, in brackets: [He] [Ne] [Ar] [Kr] [Xe] [Rn]. Then write the rest of the row in filling order.